# MG3DSCSTA202 – DATA ANALYSIS IN INFERENTIAL STATISTICS USING R/PYTHON

## Part A

Short Answer Type Questions. Each Question Carries 2 marks.

1. Formula of Conditional Probability in terms of joint and marginal probabilities- 1 mark each.

2. Given

$$f(x,y) = \frac{xy^2}{30}, \qquad x = 1,2,3, \quad y = 1,2.$$

$$f_Y(y) = \sum_{x=1}^{3} f(x,y) = \frac{y^2}{30}(1+2+3) = \frac{y^2}{5}, \quad y = 1,2.$$

3. If

$$X_i \sim N(\mu, \sigma^2),$$

then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

and

$$f_{\bar{X}}(x) = \frac{\sqrt{n}}{\sigma\sqrt{2\pi}} \exp\left[-\frac{n(x-\mu)^2}{2\sigma^2}\right].$$

4. Since

$$t_{0.025,14} = 2.262,$$

$$P(t > 2.262) = 0.025.$$

5. Applications of $F$-Distribution

(i) Testing equality of two variances,
(ii) Analysis of Variance (ANOVA).

6. 99% Confidence Level A 99% confidence level means that, in repeated sampling, approximately 99% of intervals constructed by the same method will contain the true population parameter.

7. Properties of a Good Point Estimator

(i) Unbiasedness,
(ii) Consistency,
(iii) Efficiency,
(iv) Sufficiency.

8. Type II error occurs when $H_0$ is not rejected although it is false.

9. Assumptions of $t$-Test. Any two assumptions give 2 marks.

(i) Independent observations,
(ii) Normally distributed population,
(iii) Continuous observations,
(iv) Equal variances for the pooled two-sample test.

10. One-way means that only one factor is considered.

One-way ANOVA involves one factor with two or more levels.

# Part B

Short Essay Questions. Answer any 4 questions. Each question carries 6 marks.

11. Given

$$f(x,y) = cxy, \qquad x = 0, 1, \quad y = 1, 2.$$

**(i) Find** $c$

$$\sum_x \sum_y cxy = 1$$

$$c(1)(1) + c(1)(2) = 1$$

$$c = \frac{1}{3}.$$

**(ii) Find** $P(X = 1, Y = 2)$

$$P(X = 1, Y = 2) = f(1, 2) = 2c = \frac{2}{3}.$$

**(iii) Find** $P(Y = 2)$

$$P(Y = 2) = f(0, 2) + f(1, 2) = \frac{2}{3}.$$

2 marks each.

12. If

$$Z_i \sim N(0, 1),$$

then

$$\sum_{i=1}^{n} Z_i^2 \sim \chi_n^2.$$

If

$$Z \sim N(0, 1), \quad V \sim \chi_\nu^2,$$

independently, then

$$\frac{Z}{\sqrt{V/\nu}} \sim t_\nu.$$

If

$$U \sim \chi^2_{v_1}, \quad V \sim \chi^2_{v_2},$$

independently, then

$$\frac{U/v_1}{V/v_2} \sim F_{v_1,v_2}.$$

13. Sampling Distribution with any one example...3+3

14. Definition...3 marks
Three important properties... 3 marks

15. Given

$$n = 100, \quad \sigma = 10, \quad \bar{X} = 168.8, \quad \mu_0 = 170.$$

$$H_0 : \mu = 170, \qquad H_1 : \mu \neq 170.$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{168.8 - 170}{10/\sqrt{100}} =$$

At $\alpha = 0.05$,

$$|Z| = 1.2 < 1.96.$$

Hence, do not reject $H_0$.

The assumption $\mu = 170$ cm is not rejected.

16. Given

$$n_1 = 222, \quad \hat{p}_1 = 0.21, \qquad n_2 = 208, \quad \hat{p}_2 = 0.18.$$

Hypotheses....1 mark

$$H_0 : p_1 = p_2, \qquad H_1 : p_1 > p_2.$$

The pooled estimate....2 marks

$$\hat{p} = \frac{222(0.21) + 208(0.18)}{430} = 0.1955.$$

Hence,

$$SE = \sqrt{0.1955(0.8045)\left(\frac{1}{222} + \frac{1}{208}\right)} \approx 0.0382.$$

.....1 mark
Therefore,

$$Z = \frac{0.21 - 0.18}{0.0382} \approx 0.785.$$

At $\alpha = 0.05$,

$$Z_{0.05} = 1.645.$$

.....1 mark
Since

$$0.785 < 1.645,$$

do not reject $H_0$.
.....1 mark

# Part C

Essay Type Questions. Answer 1 question. Each question carries 12 marks.

17. The joint distribution function....4 marks

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y).$$

Table ....4 marks

| Outcome | $X$ | $Y$ |
|---|---|---|
| $HHH$ | 2 | 3 |
| $HHT$ | 2 | 2 |
| $HTH$ | 1 | 2 |
| $HTT$ | 1 | 1 |
| $THH$ | 1 | 2 |
| $THT$ | 1 | 1 |
| $TTH$ | 0 | 1 |
| $TTT$ | 0 | 0 |

Rest.....4 marks.
18. Chi-Square Test for Homogeneity...3 marks Observed frequencies:...2 marks

| | Support | Neutral | Oppose | Total |
|---|---|---|---|---|
| $A$ | 40 | 30 | 30 | 100 |
| $B$ | 50 | 20 | 30 | 100 |
| $C$ | 30 | 40 | 30 | 100 |
| Total | 120 | 90 | 90 | 300 |

Hypotheses:...1 mark

$H_0$ : The cities are homogeneous, $H_1$ : The cities are not homogeneous.

Expected frequencies....2 marks

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{\text{grand total}},$$

giving

| | Support | Neutral | Oppose |
|---|---|---|---|
| $A$ | 40 | 30 | 30 |
| $B$ | 40 | 30 | 30 |
| $C$ | 40 | 30 | 30 |

Thus,

$$\begin{aligned}\chi^2 &= \frac{(50-40)^2}{40} + \frac{(20-30)^2}{30} \\ &\quad + \frac{(30-40)^2}{40} + \frac{(40-30)^2}{30} \\ &= 11.667.\end{aligned}$$

Degrees of freedom...1 mark

$$df = (3-1)(3-1) = 4.$$

At $\alpha = 0.05$,

$$\chi^2_{0.05,4} = 9.488.$$

Since

$$11.667 > 9.488,$$

reject $H_0$....3 marks

The opinions of the three cities are not homogeneous.